Investigating neuroreceptor-weighted fMRI signals across natural arousal state transitions

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BACKGROUND

Neurochemical regulation of transitions between arousal states has not been fully explored in humans.

This study investigates how brain signals from dopamine and norepinephrine-dense regions fluctuate during behavioral arousal transitions. To this end, we integrated fMRI and behavioral data with publicly available, PET-derived, receptor density maps to isolate neuromodulatory contributions to the BOLD signal.

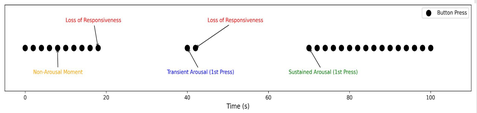
Understanding these dynamics may inform treatment of disorders involving both arousal state dysregulation and neuromodulatory imbalance (e.g. Parkinson's disease, depression)

EXPERIMENTAL DESIGN

- **Participants**: N = 21 healthy adults
- Eyes closed resting state scan
- 90 minutes of simultaneous data:
  - **fMRI** (TR = 2.22s)
  - **MR-compatible EEG**
  - **Button Presses**: Participants pressed a button with each breath

Arousal event classification:

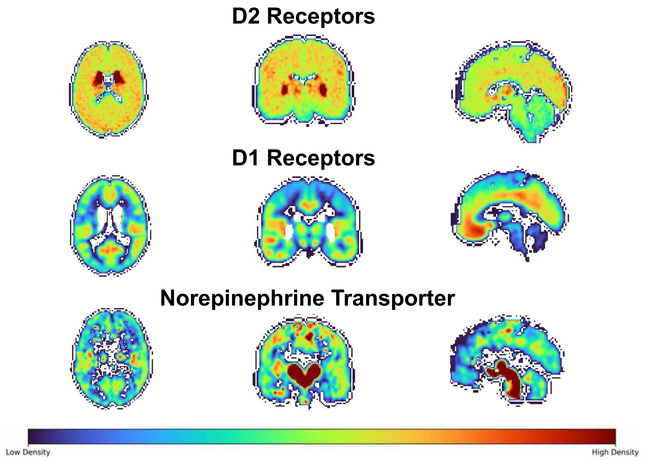
- **Loss of Responsiveness**:
  - last button press before **≥ 20s of no pressing.**
  - 308 total | 14.7 ± 9.2 per participant
- **Sustained**:
  - first press following ≥20s unresponsiveness, followed by **at least 5 consecutive responses**.
  - 116 total | 5.5 ± 4.0 per participant
- **Transient**:
  - first press following a ≥20s lapse, but with pressing continuing for **2 or fewer consecutive responses**.
  - 129 total | 6.1 ± 5.8 per participant
- **Non-Arousal Moments:**
  - button presses that occurred during continuous periods of responsiveness, (i.e.at least 8 button presses within preceding and following 20 s windows).



METHODS

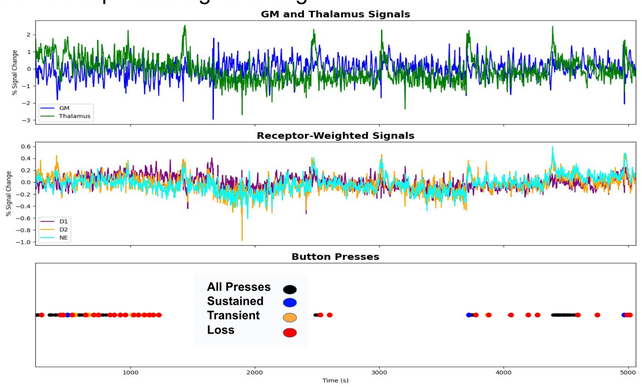
Neuroreceptor maps: PET-derived dopamine D1 and D2 receptor and norepinephrine (NE) transporter density maps obtained from Neuromaps package (Markello et al, 2021)

- Resampled to match fMRI resolution
- D1 map includes only non-striatal regions



Signal extraction:

- Sliding window regression to remove the global gray matter signal from all voxels
- Compute weighted average time series from top 90% voxels per map
- Convert to percent signal change



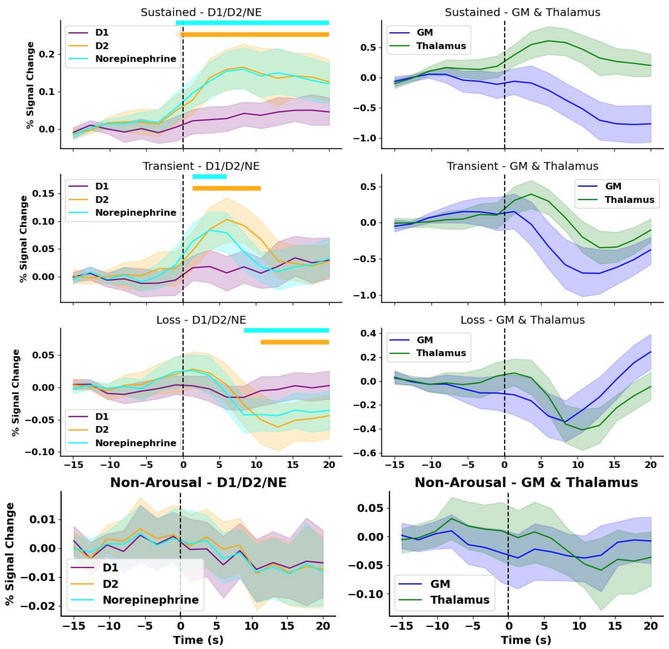
Arousal aligned BOLD responses:

- Extract fMRI signals around event onset (−15s to +20s)
- Subtract pre-event baseline [-15s,-10s]
- Hierarchical bootstrapping for 95% confidence intervals

Linear Mixed Effect Modeling:

- Estimate how arousal type and signal type affect % signal change at each time point with random intercept for each participant
  - Signal Intensity = Transient_v_Sustained + Signal_Type + (1|Participant)
  - Signal Intensity = Loss_v_Arousal + Signal_Type + (1|Participant)
- Likelihood ratio tests (LRTs) to compare nested models
- Permutation based Family Wise Error Rate control (FWER)

RESULTS



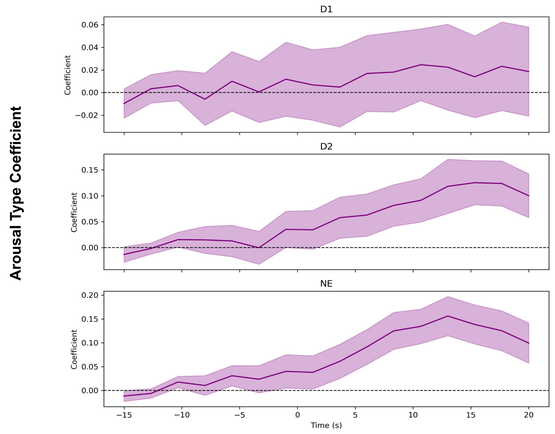
GM and Thalamus signals follow previously reported patterns across arousal state transitions (Setzer et. al., 2021).

D2 and NE signals increased significantly after arousal onset and decreased following loss of behavioral responsiveness. The increases persisted longer for sustained compared to transient arousals.

D1 signals did not show significant changes in signal amplitude locked to behavioral arousal events.

Effect of Arousal Type on Signal

Significant effect of arousal type (sustained > transient) for D2 and NE signals beginning ~2s post-arousal



Significance testing with Likelihood Ratio Tests:

We compared nested models to determine if adding a factor significantly improved model predictions. Arousal type (both transient vs sustained and arousal vs loss) significantly improved the fit for later time points ($p<0.001$).

Signal type was also a significant factor in the model at later time points ($p < 0.001$).

DISCUSSION

- Dopamine D2 and norepinephrine density-weighted BOLD signals **robustly increase** during behavioral arousal transitions. These signal changes are **more prolonged** during **sustained** arousals compared to **transient** arousals. **Loss of responsiveness** is characterized by a subtle increase beginning a few seconds before the last button press, followed by a decrease in amplitude. No changes were observed around random button presses.
- Linear mixed-effects modeling confirmed that **arousal type** and **signal type** both significantly affect fMRI signal amplitudes around arousal state transitions, with **D2 and NE**, but not D1, showing sensitivity to event type.
- The **absence of D1 effects** could reflect regional specificity or receptor affinity differences that are not detectable with the resolution of PET-derived weights.
- **Next Steps:** We will explore further the cortical and subcortical contributions to these receptor weighted fMRI signal changes around behavioral arousal transitions. We have also been investigating these fMRI signals locked to neural (EEG-based) markers of micro-arousals or sleep stages transitions on a group level.

References

Markello, R.D., Hansen, J.Y., Liu, ZQ. *et al.* neuromaps: structural and functional interpretation of brain maps. *Nat Methods* 19, 1472–1479 (2022)

Setzer, B., Fultz, N.E., Gomez, D.E.P. *et al.* A temporal sequence of thalamic activity unfolds at transitions in behavioral arousal state. *Nat Commun* 13, 5442 (2022)